

DSAA3073: Theories in Data Science

Week 1 Validation Sheet · From Uncertainty to Geometry

Coverage: Week 1

Duration: 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

Part A: Multiple choice

A1 [5 marks]. Markov's inequality can be applied directly to which variable?

- (a) Any variable with mean zero.
- (b) Any bounded variable.
- (c) A nonnegative variable with a finite mean.
- (d) Only a Gaussian variable.

A2 [5 marks]. For fixed confidence level δ , the Week 1 Hoeffding sample-size formula is proportional to $\frac{1}{\varepsilon^2}$. If ε is halved, the sufficient sample size is multiplied by

- (a) 2.
- (b) 4.
- (c) $\frac{1}{2}$.
- (d) $\frac{1}{4}$.

A3 [5 marks]. For $x = (-2, 1, 2)$, which pair is $(\|x\|_2, \|x\|_\infty)$?

- (a) (5, 2)
- (b) (3, 2)
- (c) (3, 1)
- (d) (9, 2)

A4 [5 marks]. Which statement distinguishes a Hilbert space from an arbitrary inner-product space?

- (a) Every pair of vectors is orthogonal.
- (b) Every Cauchy sequence in the induced norm converges to an element of the space.
- (c) The space must have finitely many coordinates.
- (d) Every vector has norm one.

Part B: Definition in use

B1 [20 marks]. Independent experimental units produce identically distributed pairs (A_i, B_i) , $i = 1, \dots, n$. Different pairs are independent, but A_i and B_i within the same pair may be dependent. Each coordinate lies in $[0, 1]$. The target is $\Delta = E[A_i - B_i]$, estimated by

$$\hat{\Delta}_n = \frac{1}{n} \sum_{i=1}^n (A_i - B_i).$$

Derive a valid two-sided exponential guarantee for $|\hat{\Delta}_n - \Delta|$ and a sufficient sample size for confidence $1 - \delta$. Your reasoning must make clear why the possible within-pair dependence does not invalidate the guarantee you use.

Answer space

Part C: Basic skills

C1 [7 marks]. A nonnegative random variable satisfies $E[X] = 4$. Use Markov's inequality to bound $P(X \geq 20)$.

Answer space

C2 [8 marks]. For $u = (1, 2, -2)$ and $v = (2, -1, 2)$, compute $\langle u, v \rangle$, $\|u\|_2$, and $\|v\|_2$.

Answer space

Part D: Construction and counterexample

D1 [17 marks]. Construct a nonzero vector $y = (a, b, c)$ that is orthogonal to both $u = (1, 1, 0)$ and $v = (0, 1, 1)$. Verify both inner products. Then decide whether either inequality $|\langle y, u \rangle| \leq \|y\|_2 \|u\|_2$ or $|\langle y, v \rangle| \leq \|y\|_2 \|v\|_2$ is an equality, and justify your decision.

Answer space

D2 [18 marks]. Define $q : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$q(x) = |x_1 + x_2| + |x_1 - x_2|.$$

Decide whether q is a norm. Give a complete justification of your decision; checking a few numerical vectors is not sufficient.

Answer space

Part E: Transfer through a new representation

E1 [25 marks]. Two independent instruments estimate the same mean μ . Instrument A supplies n independent readings $A_i \in [0, 1]$; instrument B supplies n independent readings $B_i \in [-1, 1]$. All $2n$ readings are mutually independent. For $0 \leq \alpha \leq 1$, consider

$$\hat{\mu}_\alpha = \alpha \left(\frac{1}{n} \sum_i A_i \right) + (1 - \alpha) \left(\frac{1}{n} \sum_i B_i \right).$$

Among all choices of α , determine the one giving the strongest two-sided exponential guarantee obtainable from the Week 1 assumptions for $|\hat{\mu}_\alpha - \mu|$. Give that guarantee and prove that no other $\alpha \in [0, 1]$ gives a better exponent by this route.

Answer space

Part F: Comparison and limitation

F1 [15 marks]. A lab records one Bernoulli variable U with $P(U = 1) = P(U = 0) = \frac{1}{2}$, then enters the same observed value into its spreadsheet n times: $X_1 = \dots = X_n = U$. An analyst argues that the average of these bounded entries has deviation probability at most $2 \exp(-2n\varepsilon^2)$.

Audit the claim. Give explicit n and ε for which the claimed inequality is false, and identify precisely what the spreadsheet failed to create despite containing n bounded entries.

Answer space